GridIQ

Site assessment: overview

Demo Corporation · Sandton Mall

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment
GQ-A-0005 · 21-day assessment

Report
revision 0 · provisional preview

Observation period (SAST)
2026-11-01 21:00 -> 2026-11-11 21:11

Observed
10.01 days of 21

Site address
99 Retail St, Sandton

Measurement points
HVAC Feed (incomer)

1. Executive findings

- Act (HVAC Feed, confidence medium; for: electrical engineer): Voltage THD on phase A exceeded 8 % (95th percentile 9.5 %), worst around 13:00, 14:00, 15:00. Individual harmonic orders were not recorded, so the source cannot be identified from this data.

2. Installation and measurement boundary

Point	Role	Nominal	CT primary	Contracted MDI	Device · installed (SAST)
HVAC Feed	incomer	230 V	–	1 400 kVA	device · 2026-08-05 22:45

Point	Coverage	Samples	Reporting interval	Gaps >= 30 min	Meter
HVAC Feed	100.0 %	15 713	300 s	2	device

Coverage = samples received × the meter's median reporting interval ÷ the observation window. Missing data is never filled or shown as zero.

sum of 1 top-level incomer(s); sub-meters are inside these and are not added

3. Energy, demand and operating pattern

HVAC Feed

Energy imported
109 361,8 kWh

Energy exported
0 kWh

Maximum billed demand (30 min)
968,1 kVA at 2026-11-06 15:00

Contracted MDI
1 400 kVA (69 % used)

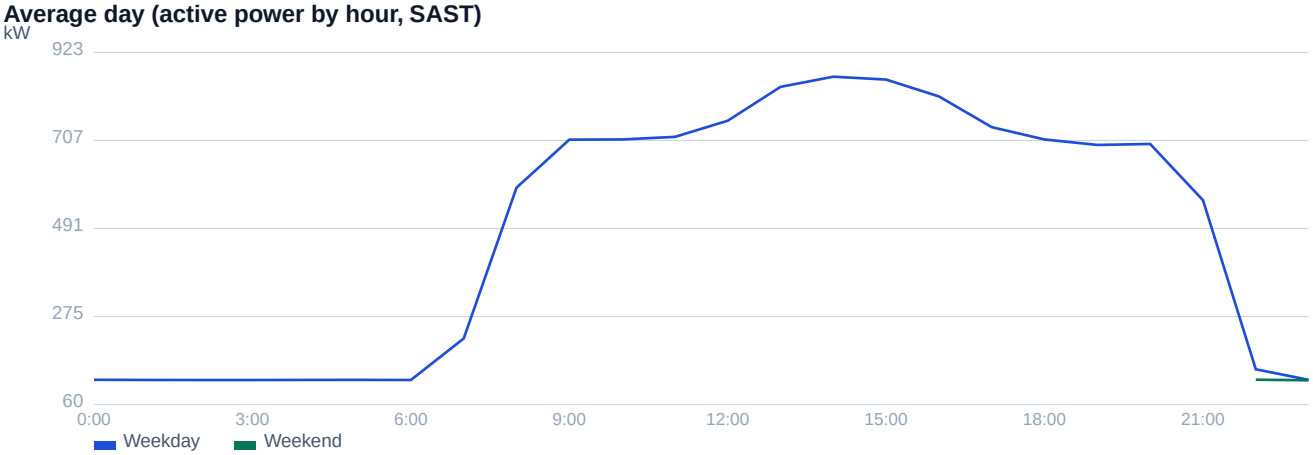
Highest instantaneous active power
955,6 kW (minute of 2026-11-06 15:28)

Baseload (10th percentile)
119,2 kW

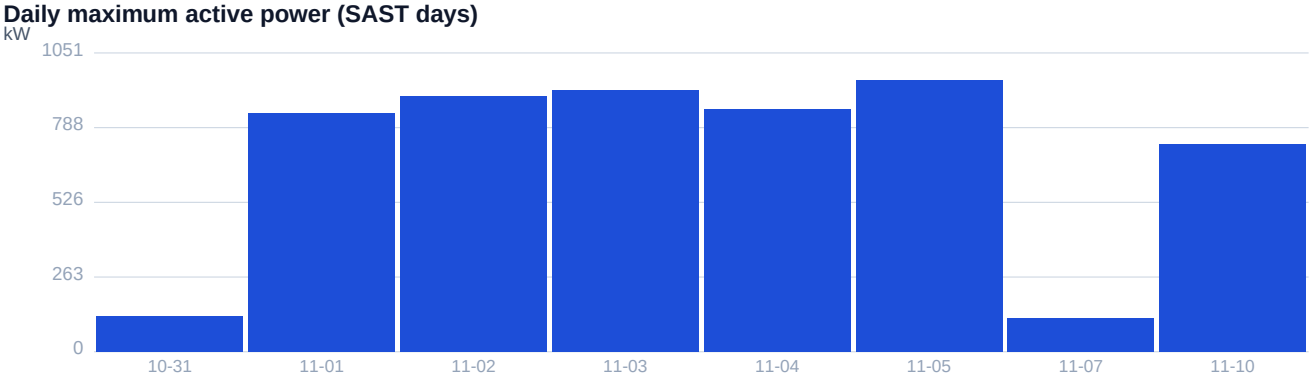
Operating-hours average
661,1 kW

Outside operating hours
277,8 kW

Average day (active power by hour, SAST)



Daily maximum active power (SAST days)



hourly averages, SAST; operating hours assumed 6:00–18:00 weekdays. Instantaneous peaks are shown separately from billed (30-minute) demand.

4. Electrical summary

- HVAC Feed: voltage A 224,1–234,2 V, B 223,9–234,1 V, C 225,2–234,5 V; THD-V p95 A 9.5 %, B 9.3 %, C 9.7 %; PF (loaded) 0.933; 0 minute(s) below 23 V.

Screening against NRS 048-2 thresholds; see the power-quality report for detail and the basis of every figure.

5. Notable events and data limitations

- PROVISIONAL: the assessment is still running; these results cover the elapsed period only and are not final.
- Observation period 10 days (2026-11-01T19:00:36.256Z to 2026-11-11T19:11:35.587Z); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.
- HVAC Feed: the meter's capability profile is not confirmed; only channels present in the data are reported.

0 voltage/frequency excursion event(s) recorded; inferred from the sampled RMS trend (pq_events), not device-recorded.

6. Prioritised actions

#	Action	Who investigates	Urgency	Confidence
1	Voltage THD on phase A exceeded 8 % (95th percentile 9.5 %), worst around 13:00, 14:00, 15:00. Individual harmonic orders were not recorded, so the source cannot be identified from this data.	electrical engineer	within a month	medium

7. Feasibility-data readiness

Supplied: 30-minute interval load profile (kW, kVA); daily energy; billed maximum demand with timestamps; power-quality screening results; metering topology and observation limits.

Missing inputs: declared operating hours (default 06:00–18:00 weekdays used); supply details and tariff; existing PV / generator / battery details; fault level at the point of supply; a full month or more of data (seasonal and monthly patterns).

GridIQ supplies measured data; system sizing and financial modelling belong to the feasibility study.

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
549679499c9ac90a0a65451b344b88f1…

Calculation version
gridiq-calc-2026.11.1

Rule version
gridiq-rules-2026.11.1

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-11 21:11

Generated (SAST)
2026-11-11 21:11

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 549679499c9ac90a0a65451b344b88f14b30c0bd16e876f6f029b7a1bd2f5508